

Migration in the Face of Climate Change: Assessing the Potential of UPG Programs

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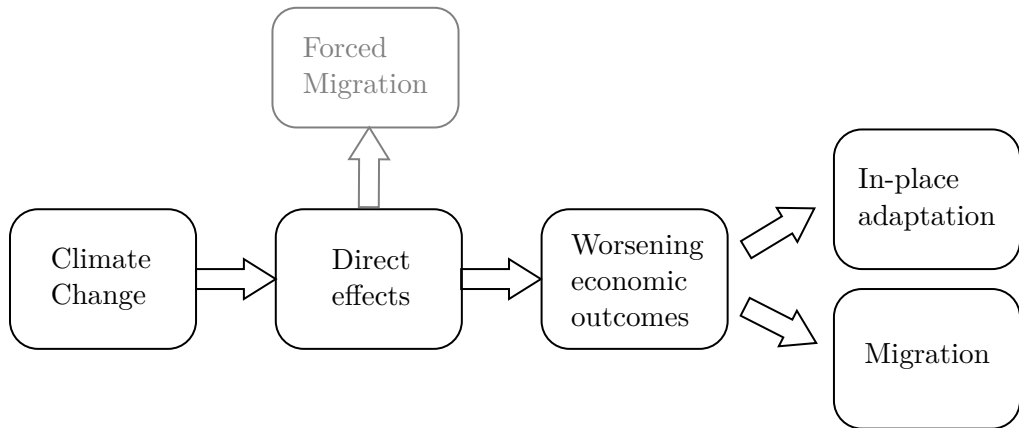
Climate Vulnerability in the Developing World

- ▶ The effects of **climate change** already are, and will continue to be, concentrated in developing countries (Desmet and Rossi-Hansberg, 2015; Costinot et al., 2016)
- ▶ Ultra-poor agrarian communities are **disproportionately exposed and at risk**
- ▶ Migration is a **potential adaptation strategy**, offering vulnerable households:
 - Less risky income sources (Dell et al., 2014; Hirvonen, 2016)
 - Protection from the labor productivity and health impacts of climate stress (Graff Zivin and Neidell, 2014; R. Burgess et al., 2017)
 - Less dependency on local markets sensitive to climate shocks (Jessoe et al., 2018)

This Project

1. How do in-place policies that target poverty alleviation shape households' capacity to adapt to climate change in-situ vs. through migration?
*⇒ In-place policies impact **exposure to climate stress**; we empirically test how exposure affects ultra-poor households' migration decisions, plans and preferences*
2. Which beneficiary households will eventually migrate away from climate-stressed areas, what will this migration look like, and who will be left behind? What features of in-place poverty-alleviation programs drive these patterns?
*⇒ Leverage bespoke survey data to **structurally estimate a dynamic discrete-choice model** to study household migration trajectories*

Theory of Change



We focus on **slow onset climate change** (ie: heat stress and drought).

Context – Upper Egypt



Setting - Ultra-Poor Graduation Program

- ▶ **Multi-faceted poverty alleviation program:** Productive asset transfer paired with technical training, plus coaching, access to savings, and consumption support.
- ▶ **Productive Asset Choice:** 80% of households chose livestock
- ▶ **Treatment arms:**
 - **Full-Cost** = asset (worth EGP 11,000) + intensive coaching/training
 - **Half-Cost** = 1/2 asset value + version lower-intensity coaching/training
- ▶ **Eligibility:** Proxy means testing to target ultra-poor households with additional exclusion criteria: (i) no able-bodied working-aged member, (ii) ownership of > 0.42 hectares (0.5 feddan) of land, or (iii) $> \text{EGP } 12,000$ annual remittances.

Sample

- ▶ **3,469 ultra-poor rural households** across Assuit and Sohag
 - Full sample from an ongoing RCT of the UPG Program in Upper Egypt
 - **Three treatment arms** – full-value (50%), half-value (25%) and control (25%)
 - **Surveys:** baseline (2019), midline (2021), endline I (2023) and **endline II (ongoing)**
- ▶ Baseline profile about a typical HH
 1. **HH Size** - 7 individuals in median household (3 males, 3 females, 1 child)
 2. **Typical Occupations** - agriculture, street vendors/merchants, construction
 3. **Livestock** - 50% own livestock, 20% own "big" livestock (sheep, cow, buffalo etc.)

Our Climate Migration Survey (Five Years After UPG)

- ▶ Up to three surveys per HH – main female respondent and 1 or 2 working-aged males. We will conduct a total of 8,016 surveys across Assuit and Sohag.
- ▶ Elicit patterns, plans and desires on *different types* of migration –
 - **Male survey** – individual's past trips, plans and desires for migrating
 - **Female survey** – fill in gaps for men currently migrating
- ▶ Additional modules on livestock, climate beliefs, water use, occupation, productivity, and crops to pin down exposure to climate stress
- ▶ Staggered treatment rollout $\Rightarrow$ four batches surveyed at different times throughout the year, allowing us to capture seasonal migration trends
- ▶ Will match to previous UPG survey waves for income, expenditure, etc.

Stylized Facts

From Climate Data:

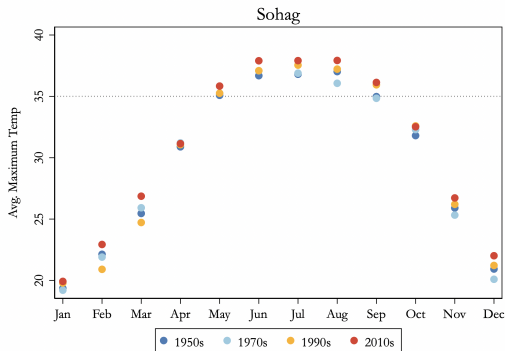
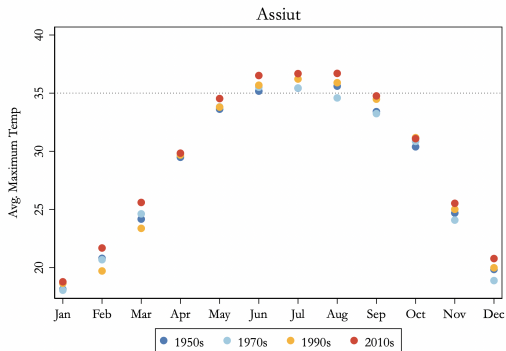
1. Climate in Upper Egypt is becoming increasingly **hotter** and **drier**

From Our Survey Data:

2. Households are **aware** and **worried** about mounting climate stress
3. Concern and stress coincide with **in-situ adaptation**
4. **Migration patterns** - who moves? how many move? where do they go?

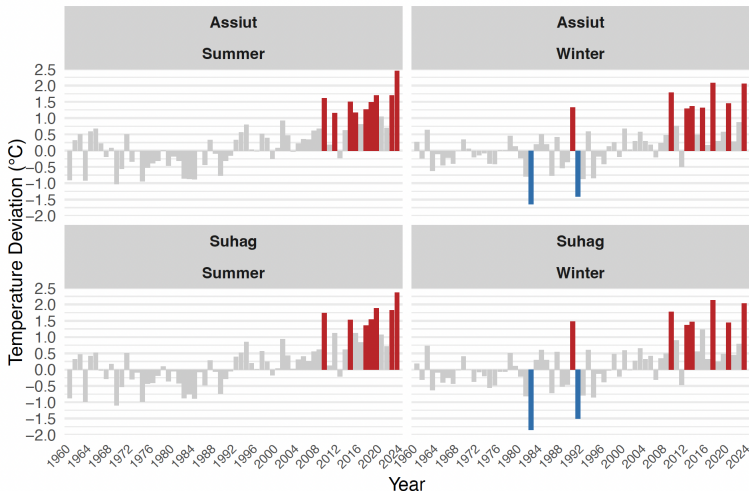
1a. Upper Egypt is getting hotter

Each summer, Assiut and Sohag average 16.7 and 24.7 days above 40 degrees.



1a. Upper Egypt is getting hotter

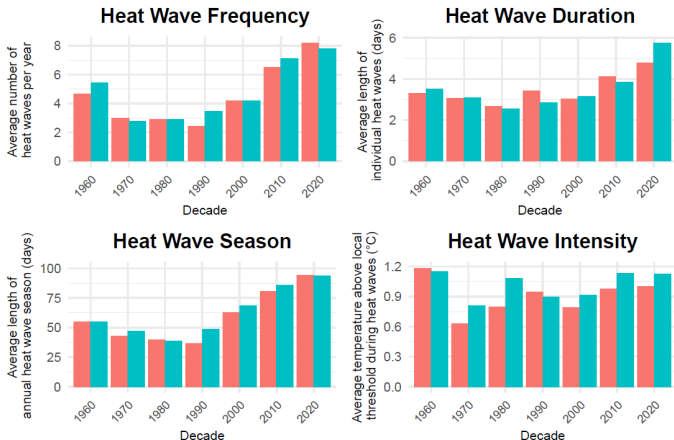
Annual Mean Temperature, relative to 1960-2010 seasonal means



1a. Upper Egypt is getting hotter

Heat Wave Incidence by Governorate

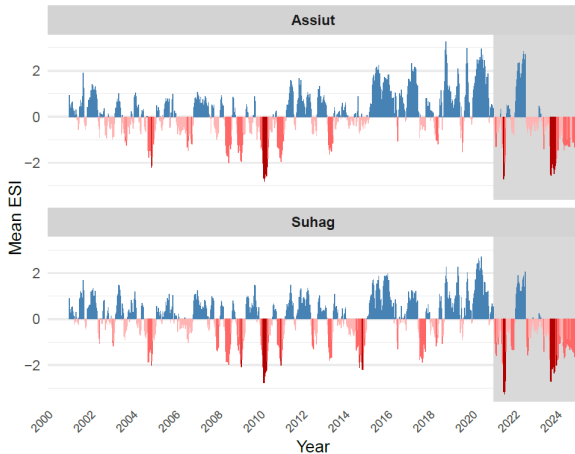
The frequency, duration, and intensity of heat waves have risen significantly. They also occur earlier and later in the year than they did historically.



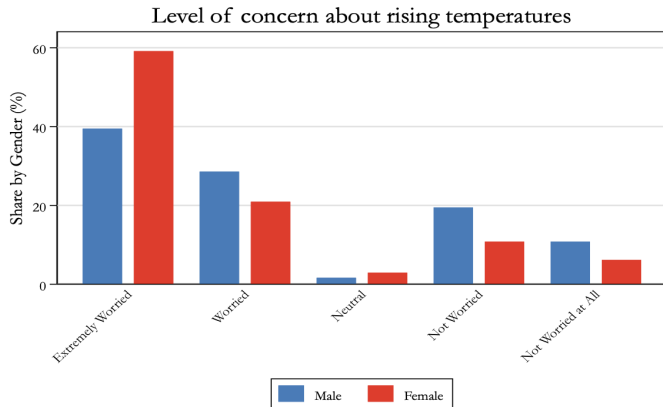
1b. Upper Egypt is getting drier

Weekly Evaporative Stress Index (**drought stress** to **wet anomalies**)

Exceptionally and long-lasting dry conditions since 2021



2a. Concern over Rising Temperatures

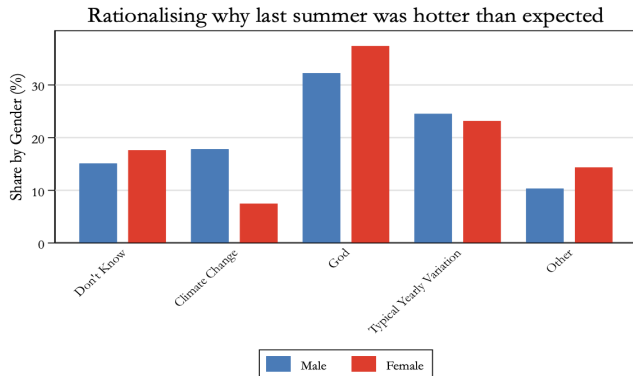


Sample -- 1221 males and 1395 females

Similar patterns for concern over: (i) effect of temperature on own and livestock health and productivity, (ii) quality/availability of livestock fodder in summer and (iii) water shortages

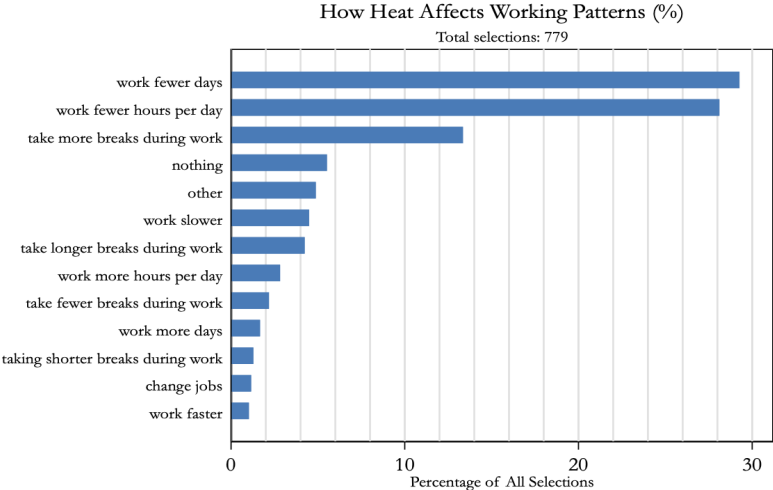
2b. Interpreting Signals as Climate Change

- ▶ 31% of the sample has heard of the phrase “climate change”
- ▶ Of these, 86% correctly identify the local consequences of climate change (rising temperatures, less predictable/more extreme seasons etc.)



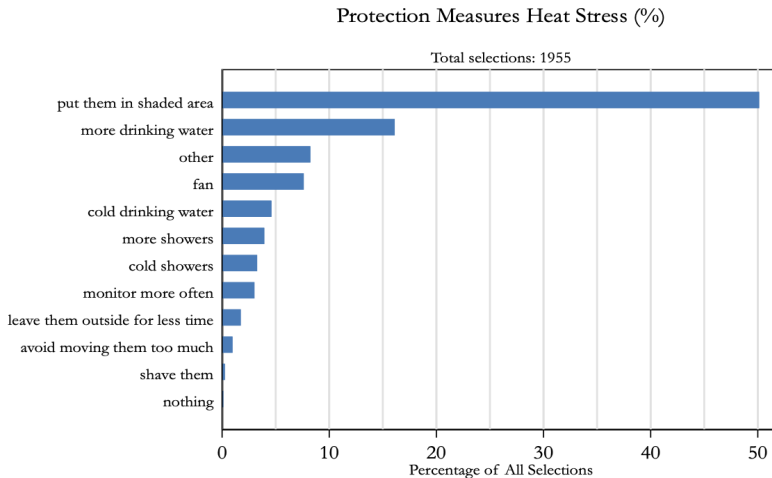
Sample -- 775 males and 976 females, conditional on thinking it was hotter than expected
73.1% of females, 66.8% of males report it being hotter than expected

3a. In-Situ Adaptation



N = 544 males, conditional on having worked in the past 30 days

3b. In-Situ Adaptation



N = 1103 HHs, conditional on a HH owning livestock

4. Migration Patterns

- ▶ 66.8% of working-age males engage in work-related temporary migration¹
 - Average of 1 trip per year with a median duration of 60 days
- ▶ 16.1% of working-age males engage in work-related long-term migration
 - On average, they spend 8 months away from the HH in a year
- ▶ Majority go to urban areas in the north of Egypt
 - Complementarities → urban wage premium + more productive in cooler climate
- ▶ Occupation distribution surprisingly comparable in origin / destination
 - 67% of migrants work in construction, agriculture or unskilled service jobs
 - 55% report having the same occupation as when they are in the village

¹Working-aged males are those between 18-60. We define temporary migration as Banerjee et al. (2021) do – more than 2 consecutive nights spent away from the HH, in this case for work.

Treatment Effects on Migration

To estimate the impact of treatment on migration, we estimate:

$$M_{i,t=5} = \beta_0 + \beta_1 FullCost_i + \beta_2 HalfCost_i + \gamma' Controls_{i,t=0} + \eta_v + \epsilon_{i,t=5},$$

where $M_{i,t=5}$ is a migration outcome captured in our five-year endline survey, $\{FullCost_i, HalfCost_i\}$ are dummies capturing household i 's treatment status of household i , $Controls_{i,t=0}$ are baseline controls and η_v are village fixed-effects.

Noisy Effects of Treatment on Migration

Table: Effect of UPG Program on Household Migration Patterns

	HH has ≥ 1 migrant	Proportion of migrants	Number of migrants	Remittance total
Full-Cost	−0.002 (0.027)	−0.012 (0.022)	0.018 (0.050)	−441.223 (368.098)
Half-Cost	−0.041 (0.030)	−0.038* (0.022)	−0.063 (0.039)	−538.728 (500.452)
Obs	3,001	3,001	3,001	3,001

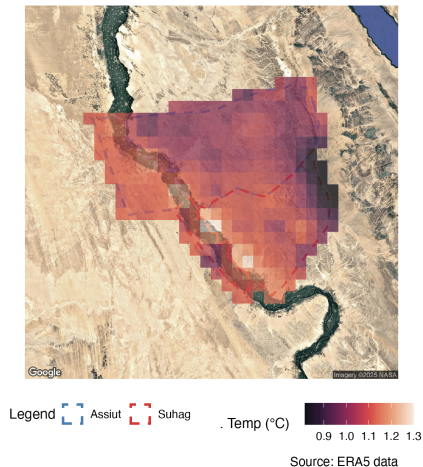
Notes: I report significance at the 1% (* * *) 5% (**), and 10% (*) levels with standard errors clustered at the village-level. Baseline controls: number of members in HH, HH head gender/age/education level, dwelling type/area/ownership/no. of rooms.

Measuring Climate Exposure

- ▶ We will construct a **novel index of exposure to climate stress**, combining:
 - *Exogenous* spatial variation in **climate shocks** and **access to water** to cope with them
 - *Endogenous* household-level variation in **occupation** and **livestock holdings**
- ▶ For now, we construct an Inverse Correlation Weighted (ICW) Index using indices for heat stress, drought (ESI) and ease of water access. We use:
 - **Spatial variation** in temperatures and ESI (means, trajectories and variance)
 - **Access to Water** - proximity and commuting time to canals
- ▶ **Eventual Goal:** Combine climate stress indices with indices that capture climate vulnerability through household occupations and livestock holdings. Then: instrument using randomized receipt of the UPG program and re-center.²

²We will simulate counterfactual climate shocks and program randomization to re-center our instrument. This adjusts for the fact that certain ultra-poor households are more likely to be exposed, and exploits randomness in realized climate exposures relative to expected levels (Borusyak and Hull, 2023)

Spatial Variation in Heat and Drought Stress



Variation in:

- ▶ Average temp./ESI
- ▶ *Changes* in mean temp. (see figure) and ESI
- ▶ Incidence and frequency of heat waves / hot days
- ▶ Incidence and frequency of shocks in the ESI
- ▶ Season start and end
- ▶ Variance in temp./ESI

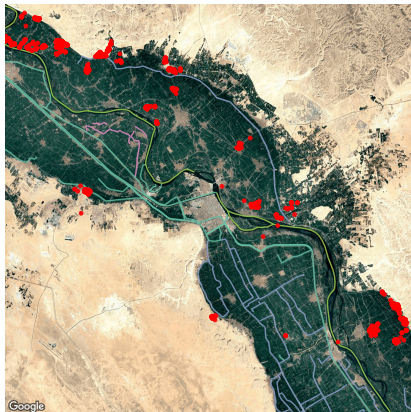
Survey Data

Fig: Variation in change in mean temp. across Upper Egypt

Access to Water $\simeq$ Proximity to Canals

Waterways & Household Locations

Households within 30 km of Asyut City

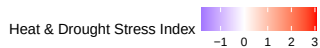
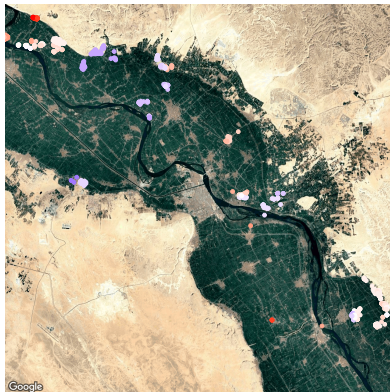


Waterway type — Canal — Ditch — Drain — River

Describing Climate Exposure

Household Heat & Drought Stress Index

Households within 30 km of Asyut City



Climate Exposure $\Rightarrow$ Migration

We estimate the following specifications:

$$M_{i,t=5} = \beta_0 + \beta_1 Exposure_{i,t=0-5} + \beta_2 Exposure_{i,t<0} + \gamma' Controls_{i,t=0} + \eta_v + \epsilon_{i,t=5}$$

$$M_{i,t=5} = \beta_0 + \beta_1 FullCost_i + \beta_2 HalfCost_i + \beta_3 FullCost_i \cdot Exposure_{i,t=0-5} \\ + \beta_4 HalfCost_i \cdot Exposure_{i,t=0-5} + \gamma' Controls_{i,t=0} + \eta_v + \epsilon_{i,t=5},$$

where $M_{i,t=5}$ is a migration outcome captured in our five-year endline survey, $Exposure_{i,t=0-5}$ is our contemporaneous climate exposure index, $Exposure_{i,t<0}$ is our historical exposure index (2000-2020) and $Controls_{i,t=0}$ are baseline controls.

Climate Stress Attenuates Migration Rates

Table: Effect of Exposure to Climate Stress on Household Migration Patterns

	HH has ≥ 1 migrant	Proportion of migrants	Number of migrants	Remittance total
Exposure _{$t=0-5$}	-0.036*** (0.010)	-0.030** (0.012)	-0.051** (0.023)	-263.146 (223.313)
Obs	3,001	3,001	3,001	3,001

Notes: We report significance at the 1% (***) 5% (**), and 10% (*) levels with standard errors clustered at the village-level. Baseline controls: number of members in HH, HH head gender/age/education level, dwelling type/area/ownership/no. of rooms. We additionally control for historical exposure (2000-2020). All exposure indices are standardised.

Treatment $\times$ Climate Exposure

Table: Effect of Exposure to Climate Stress on Household Migration Patterns

	HH has ≥ 1 migrant	Proportion of migrants	Number of migrants	Remittance total
Full-Cost	0.001 (0.026)	-0.019 (0.021)	0.021 (0.050)	-660.347 (473.919)
Half-Cost	-0.047 (0.031)	-0.040* (0.021)	-0.068* (0.034)	-718.401 (617.609)
Full-Cost $\times$ Exposure_{$t=0-5$}	-0.041*** (0.012)	-0.025* (0.013)	-0.069** (0.032)	-163.603 (286.733)
Half-Cost $\times$ Exposure_{$t=0-5$}	-0.036 (0.028)	-0.023 (0.027)	-0.052 (0.042)	-49.952 (244.270)
Obs	3,001	3,001	3,001	3,001

Notes: We report significance at the 1% (***) 5% (**), and 10% (*) levels with standard errors clustered at the village-level. Baseline controls: number of members in HH, HH head gender/age/education level, dwelling type/area/ownership/no. of rooms. We additionally control for historical exposure (2000-2020). All exposure indices are standardised.

Summary

1. **The UPG program has noisy effects on household migration patterns.**
⇒ Half-cost treatment reduces migration relative to the control group.
2. **Contemporaneous exposure to climate stress reduces out-migration**
⇒ This effect is entirely driven by HHs who live farther away from the Nile, which gives them lesser (and more unpredictable) access to water to cope.
3. **In climate-stressed areas, the full-cost treatment reduces migration**
⇒ Is this because they are vulnerable, through heat-stressed livestock, OR because they have greater ability or willingness to adapt in-situ?

Next Step: Structural Model

Embed a **dynamic discrete-choice** of the decision to migrate in **general equilibrium**

Key Ingredients:

- ▶ Rationalise different migration decisions (seasonal, temporary, permanent)
- ▶ Incorporate salience of, and beliefs surrounding, climate change
- ▶ Endogenous savings, asset holdings and occupational choice, which are chosen, jointly with migration, in response to climate shocks and UPG program receipt
- ▶ Equilibrium adjustments in local wages and asset prices (Jessee et al., 2018)

Key insights: (i) who chooses to migrate and when, (ii) who is left behind, (iii) how do alternative UPG program designs (e.g. eligibility) change these findings

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